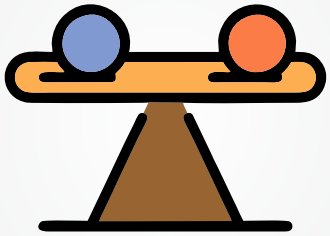
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Center of Mass, Momentum & Collision

(English Medium)

Copyright Statement

All rights including trademark and copyrights and rights of translation etc. reserved and vested exclusively with ALLEN Career Institute Private Limited. (ALLEN)

No part of this work may be copied, reproduced, adapted, abridged or translated, transcribed, transmitted, stored or distributed in any form retrieval system, computer system, photographic or other system or transmitted in any form or by any means whether electronic, magnetic, chemical or manual, mechanical, digital, optical, photocopying, recording or otherwise, or stood in any retrieval system of any nature without the written permission of the Allen Career Institute Private Limited. Any breach will entail legal action and prosecution without further notice.

This work is sold/distributed by Allen Career Institute Private Limited subject to the condition and undertaking given by the student that all proprietary rights (under the Trademark Act, 1999 and Copyright Act, 1957) of the work shall be exclusively belong to ALLEN Career Institute Private Limited. Neither the Study Materials and/or Test Series and/or the contents nor any part thereof i.e. work shall be reproduced, modify, re-publish, sub-license, upload on website, broadcast, post, transmit, disseminate, distribute, sell in market, stored in a retrieval system or transmitted in any form or by any means for reproducing or making multiple copies of it.

Any person who does any unauthorised act in relation to this work may be liable to criminal prosecution and civil claims for damages. Any violation or infringement of the propriety rights of Allen shall be punishable under Section- 29 & 52 of the Trademark Act, 1999 and under Section- 51, 58 & 63 of the Copyright Act, 1957 and any other Act applicable in India. All disputes are subjected to the exclusive jurisdiction of courts, tribunals and forums at Kota, Rajasthan only.

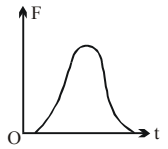
Note:- This publication is meant for educational and learning purposes. All reasonable care and diligence have been taken while editing and printing this publication. ALLEN Career Institute Private Limited shall not hold any responsibility for any error that may have inadvertently crept in.
ALLEN Career Institute Private Limited is not responsible for the consequences of any action taken on the basis of this publication.

CENTER OF MASS, MOMENTUM & COLLISION

KEY CONCEPT

The action of force with respect to time is defined in terms of Impulse, that is,

$$I = \int F dt = mv_f - mv_i = \Delta p$$



Graphically, impulse is the area under the F-t graph

In the absence of a net external force, the momentum of a system is conserved.

i.e. $\frac{dP}{dt} = F_{ext} = 0$ $\qquad p = p_1 + p_2 + ...+ p_N = \text{constant}$

1. **Collision** is a kind of interaction between two or more bodies which come in contact with each other for a very short time interval.

2. **Types of collision: Elastic and Inelastic**

Collisions may be either elastic or inelastic. Linear momentum is conserved in both cases.

(i) A perfectly elastic collision is defined as one in which the total kinetic energy of the system is conserved.

(ii) In an inelastic collision, the total kinetic energy of the system changes.

(iii) In a completely inelastic collision, the two bodies couple or stick together.

3. **Coefficient of Restitution :** It is defined as the ratio of the velocity of separation to the velocity of approach of the two colliding bodies.

$$e = \frac{\text{relative velocity of separation}}{\text{relative velocity of approach}}$$

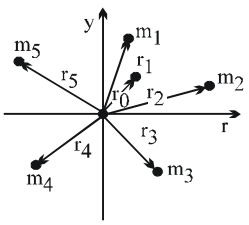
For a perfectly elastic collision, e = 1 $\qquad$ For an inelastic collision, 0 < e < 1

For completely inelastic collision, e = 0

Note that the velocity of approach and the velocity of separation are always taken along the normal to the striking surface.

CENTRE OF MASS

1. **Discrete System :** The position vector of the centre of mass is



$$r_c = \frac{m_1 r_1 + m_2 r_2 + ... + m_n r_n}{m_1 + m_2 + ...m_n}$$

where $\vec{r}_1, \vec{r}_2, ..., \vec{r}_n$ are the position vectors of masses m_1, m_2, ..., m_n respectively.

The components of the position vector of centre of mass are defined as

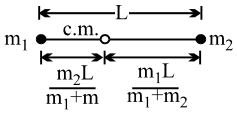
$$x_c = \frac{\sum m_i x_i}{M}; \qquad y_c = \frac{\sum m_i y_i}{M}; \qquad z_c = \frac{\sum m_i z_i}{M}$$

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll.\8 Rat Dy\Hindi\03_COM & Collision\01_Hindi.p65

2. Continuous system : The centre of mass of a continuous body is defined as $\vec{r}_c = \frac{1}{M}\int r\,dm$

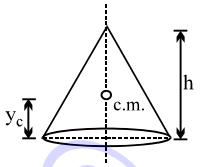
In the component form $x_c = \frac{1}{M}\int x\,dm$; $y_c = \frac{1}{M}\int y\,dm$; $z_c = \frac{1}{M}\int z\,dm$

3. Centre of Mass of Some Common Systems :

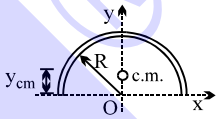


(i) A system of two point masses.
The centre of mass lie closer to the heavier mass.

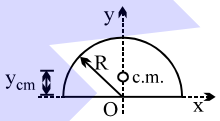
(ii) A circular cone $y_c = \frac{h}{4}$



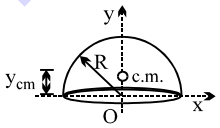
(iii) A semi-circular ring $y_c = \frac{2R}{\pi}$; $x_c = 0$



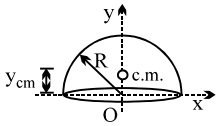
(iv) A semi-circular disc $y_c = \frac{4R}{3\pi}$; $x_c = 0$



(v) A hemispherical shell $y_c = \frac{R}{2}$; $x_c = 0$



(vi) A solid hemisphere $y_c = \frac{3R}{8}$; $x_c = 0$



4. Motion of the centre of mass :

(i) **Velocity :** The instantaneous velocity of the centre of mass is defined as $v_c = \frac{\sum m_i v_i}{M}$

(ii) **Acceleration :** The acceleration of the centre of mass is defined as $a_c = \frac{\sum m_i a_i}{M}$

(iii) **Momentum :** The total momentum of a system of particles is $p = Mv_c$

(iv) **Kinetic Energy :** The kinetic energy of a system of particles consists of two parts. $K = K_c + K'$

where $K_c = \frac{1}{2}Mv_c^2$, kinetic energy due to motion of c.m. relative to the fixed origin O and

$K' = \sum\frac{1}{2}m_i v_i^2$, kinetic energy of the particles relative to the c.m.

Note that the term K' may involve translational, rotational or vibrational energies relative to the centre of mass.

ALLEN®

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Collision & Rot Dy\Hindi\03_COM & Collision\01_Hindi.p65

5. Newton's Laws of a system of particles :

The first and second laws of motion for a system of particles are modified as :

First law : The centre of mass of an isolated system is at rest or moves with constant velocity.

Second law : The net external force acting on a system of total of mass M is related to the acceleration of centre of mass of the system.

$$\sum \vec{F}_{ext} = M\vec{a}_{cm}$$

INTEGRATION PRACTICE SHEET

Section A: Uniform Density Objects

1. A rod of length L is placed along the X-axis between x = 0 and x = L. The linear density (mass/length) ρ of the rod varies with the distance x from the origin as ρ = a + bx.

(a) Find the SI units of a and b.

(b) Find the mass of the rod in terms of a, b and L.

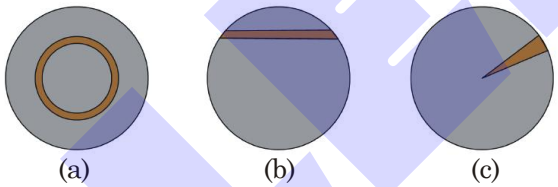
Ans. (a) kg/m, kg/m² (b) aL + bL²/2

2. The ratio of the circumference of a circle and its diameter is π. Prove that the area of a circle of radius R is A = πR². Do it by taking the 'elemental area' and integrating. The elemental area could be:

(a) A thin ring with center same as that of the circle.

(b) A thin, almost rectangular chord, at some distance less than 'R' from the center.

(c) A thin sector of the circle.



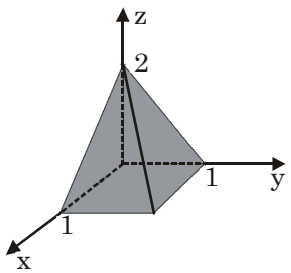
3. Derive the expression for the surface area of a hollow sphere of radius R.

Ans. $4\pi R^2$

4. Derive the expression for volume of a solid sphere of radius R.

Ans. $\frac{4}{3}\pi R^3$

5. Consider a pyramid with a one-by-one square base along the x-y plane with corners at (0,0,0), (0,1,0), (1,1,0) and (1,0,0) and tapering up to a point at (0,2,2). Find the volume of this pyramid. Notice that every cross-section parallel to the x-y plane is a square.



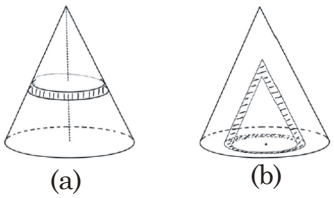
Ans. 2/3

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll.\8.Rot Dy\Hindi\03_COM & Collision\01_Hindi.p65

6. Derive the expression for the curved surface area of a hollow cone of radius R and height H.

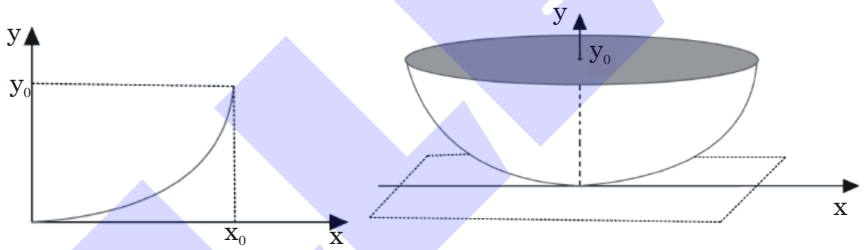
Ans. $\pi R\sqrt{R^2 + H^2}$

7. Derive the expression for the volume of a solid cone of radius R and height H. Consider the following elements, as shown in the diagram:
(a) A disk parallel to the base of the cone.
(b) A thin hollow cone with same half-angle as the solid cone.



Ans. $\pi R^2 H / 3$

8. The curve $y = kx^2$ is called a parabola. Find the area enclosed by the parabola, x-axis, and the line $y = x_0$ as shown in the first diagram below. Now the parabola is rotated about the y axis to form a cup shaped volume called a paraboloid. Find the volume of the paraboloid-cup thus formed.



Ans. $\frac{x_0 y_0}{3}, \frac{\pi x_0^2 y_0}{2}$

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot Dy\Hindi\03_COM & Collision\01_Hindi.p65